Lecture 4: AWS VPC & Routing

Maharishi International University

Department of Computer Science

M.S. Thao Huy Vu

Maharishi International University - Fairfield, Iowa

All rights reserved. No part of this slide presentation may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying or recording, or by any information storage and retrieval system, without permission in writing from Maharishi International University (MIU).

Agenda

- **Introduction to Amazon VPC**
- **VPC Core Components**
- **VPC Peering**
- **Transit Gateways**
- **VPC Flow Logs**
- **Route 53**

Introduction to Amazon VPC

- Virtual Private Cloud
- Security
- Isolation
- Flexibility

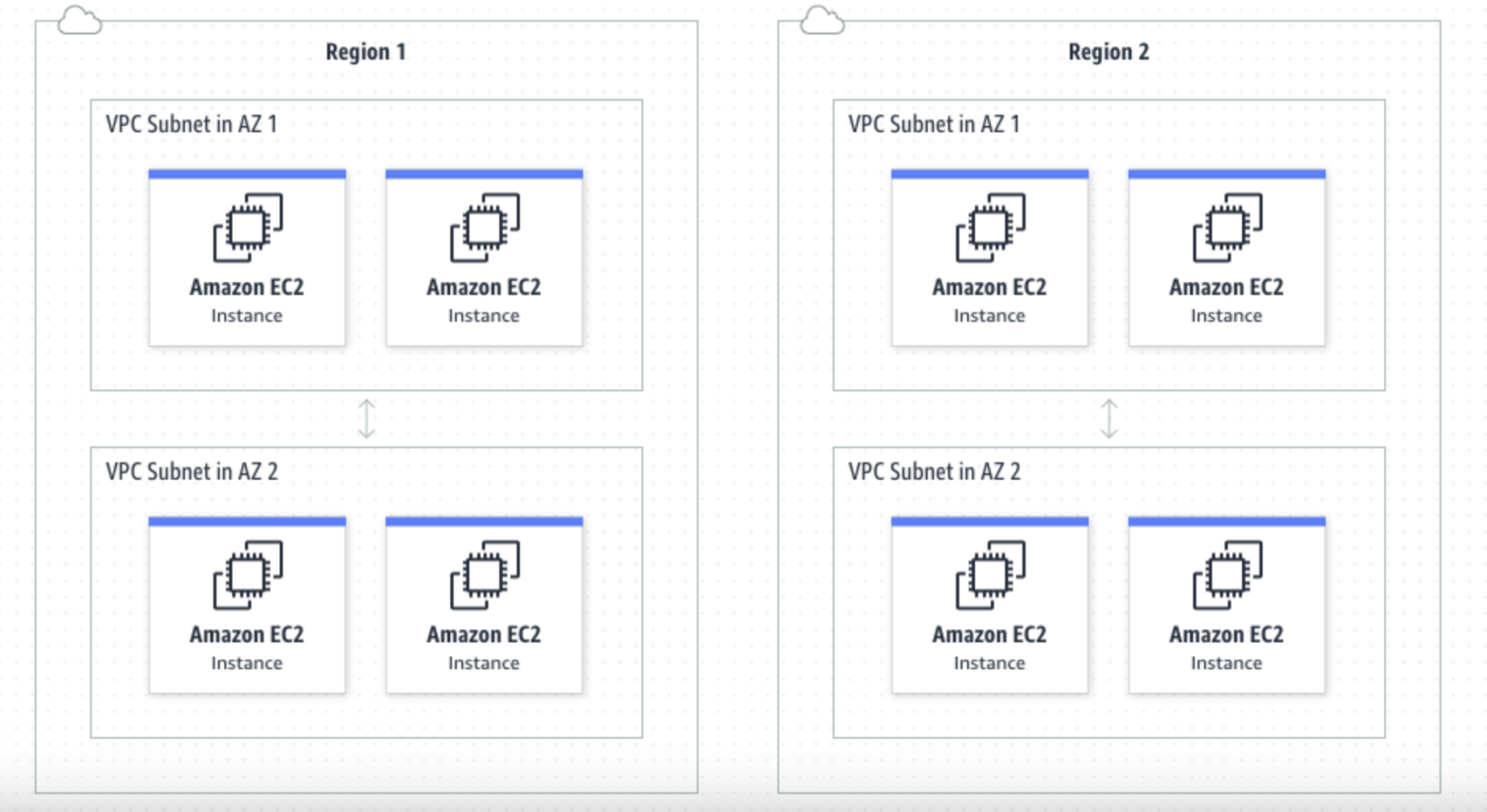
Introduction to Amazon VPC

- Core Components
  - Subnets: A VPC can be segmented into subnets, which can be designated as a public or private to control the type of access they allow to and from internet.
  - Route Tables: Rules that determine where network traffic from subnets or gateways is directed
  - Security Groups: A virtual firewall for an instance to control inbound and outbound traffic.
  - Network ACLs: An additional layer of security at the subnet level.

Two types of resources

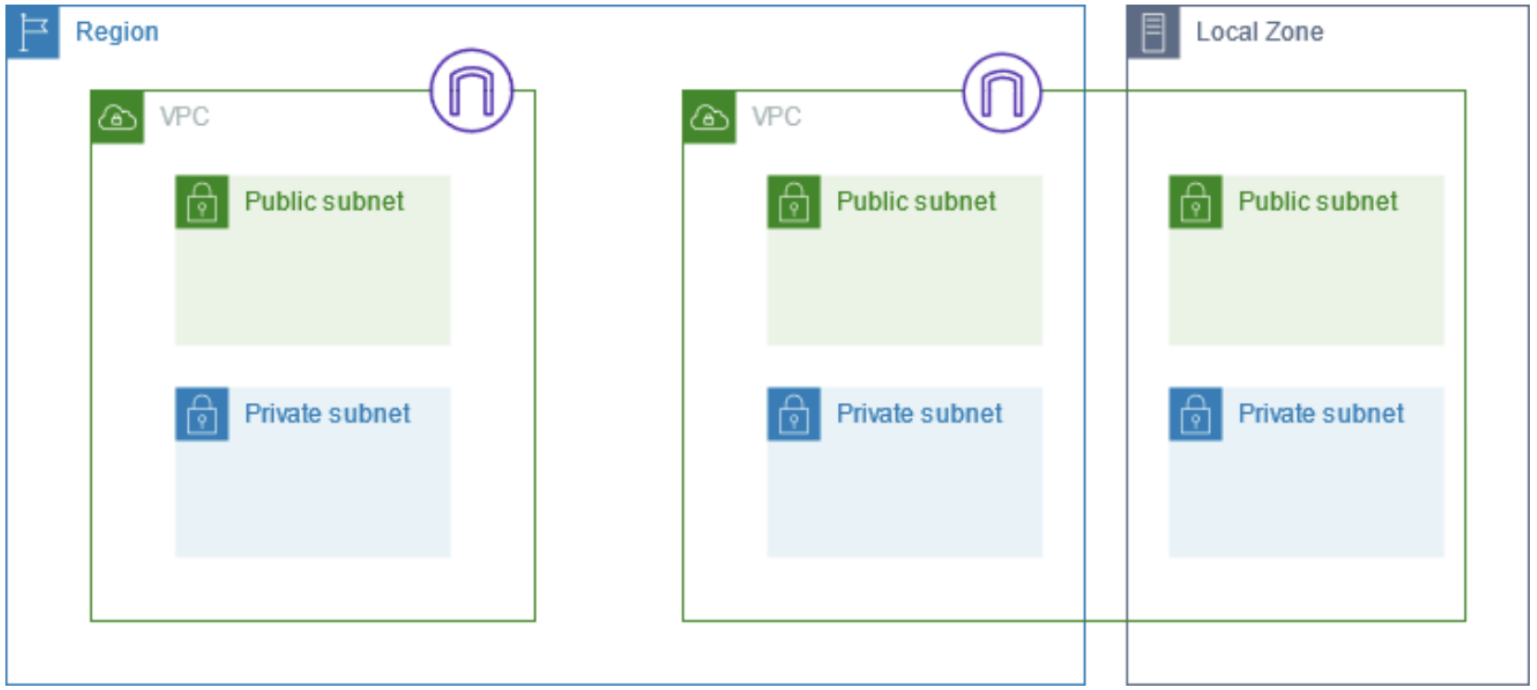
1. Public – It is just like a third-party API that you can call directly such as S3, DynamoDB, SNS, SQS, and so on (FaaS services). If valid authorization tokens are present, the API call is a success.
2. Non-public – You must launch those resources in VPC so that it gets an IP address, so others can connect to it. It includes EC2, RDS, Load Balancers, and so on (IaaS and PaaS services). It has additional security layers.

Introduction to Amazon VPC



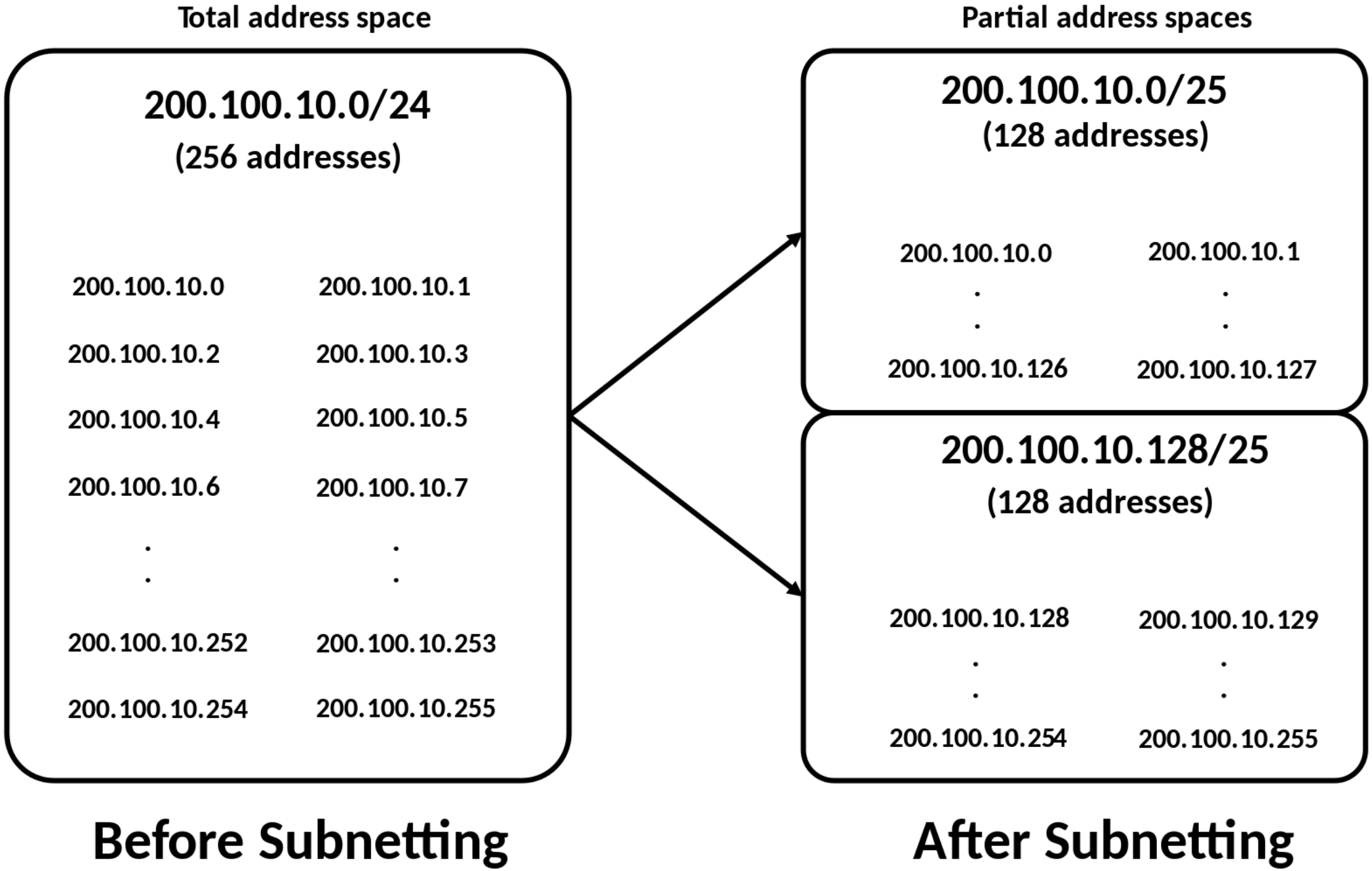
VPC Core Components

- Subnets



VPC Core Components

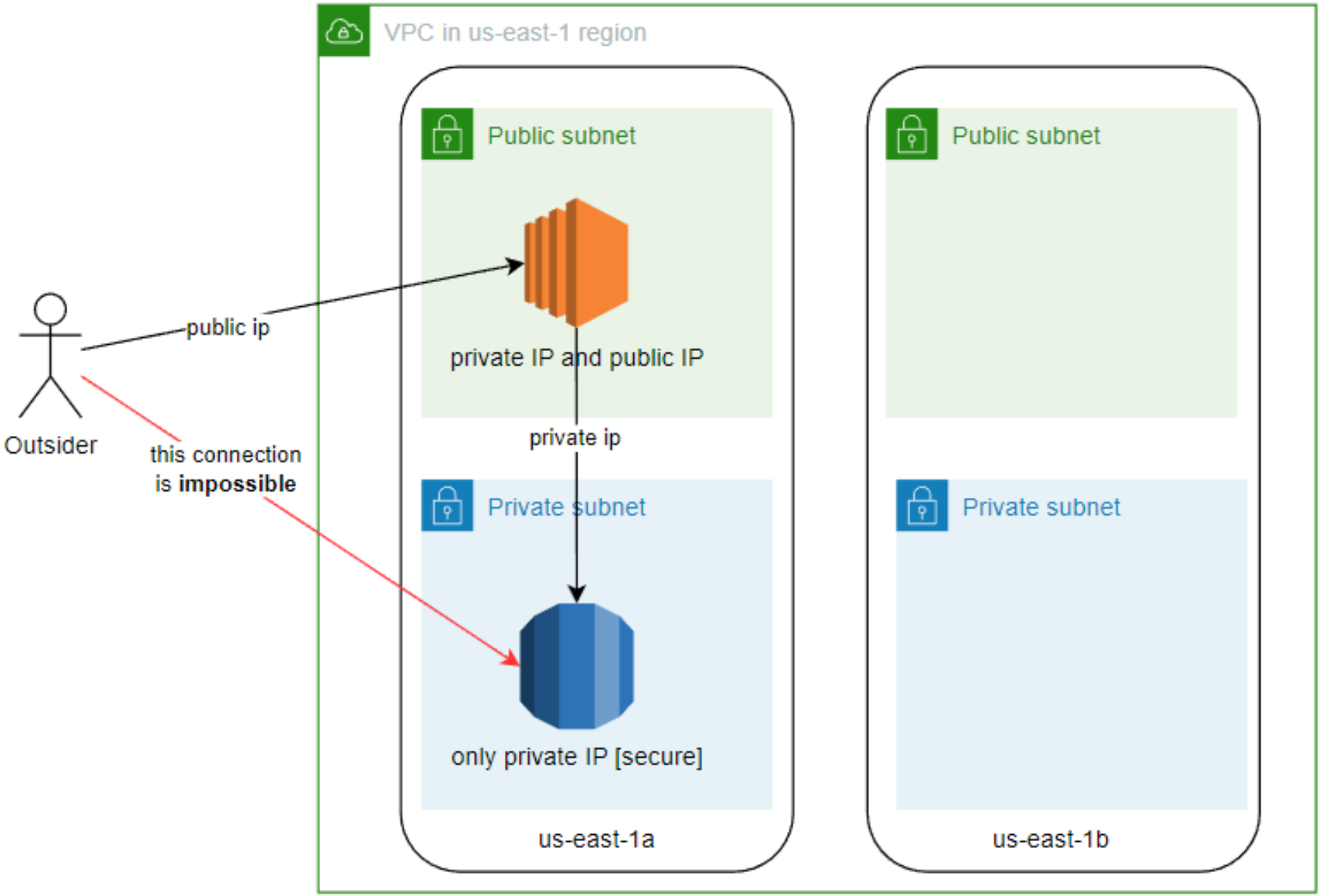
- Subnets:
  - Define a subnet as a range of IP addresses in VPC
  - Isolate different types of resources for public or private access
  - Choose the right CIDR block for subnets to efficiently utilize IP addresses
  - Associate with one route table


Total address space
200.100.10.0/24
(256 addresses)
200.100.10.0
200.100.10.1
200.100.10.2
200.100.10.3
200.100.10.4
200.100.10.5
200.100.10.6
200.100.10.7
200.100.10.252
200.100.10.253
200.100.10.254
200.100.10.255
Partial address spaces
200.100.10.0/25
(128 addresses)
200.100.10.0
200.100.10.1
200.100.10.126
200.100.10.127
200.100.10.128/25
(128 addresses)
200.100.10.128
200.100.10.129
200.100.10.254
200.100.10.255
Before Subnetting
After Subnetting

CIDR ranges for the private network

Most popular private IPs that are reserved for local network. VPC is an example of a local network.

CIDR	Starting and ending IPs	Total number of hosts
10.0.0.0/8	10.0.0.0 – 10.255.255.255	16,777,216
172.16.0.0/12	172.16.0.0 – 172.31.255.255	1,048,576
192.168.0.0/16	192.168.0.0 – 192.168.255.255	65,536



Public requests must pass the VPC security layers on the subnet and resource to hit the targeted resource.

There is no direct way to access from the public to resources in the private subnet. The only way to access private resources is through the public resources.

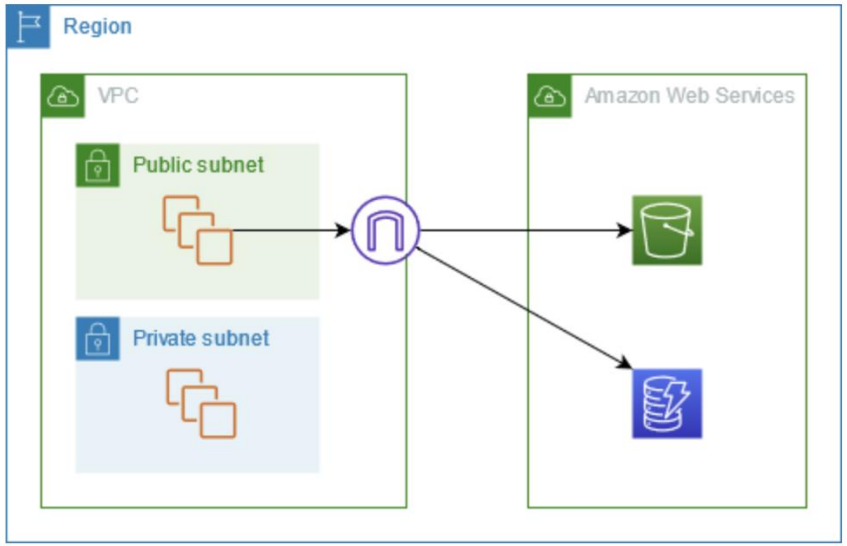
VPC Core Components

- Route Tables
  - Define rules, known as routes, which determine where network traffic from subnets or the internet is directed
  - Each subnet in a VPC must be associated with a route table, which can be the main route table or a custom table.
  - Define common routing rules, such as local routes, internet-bound routes via internet gateway, and internal routes via VPC peering

Destination	Target
10.0.0.0/16	Local
2001:db8:1234:1a00::/56	Local
172.31.0.0/16	pcx-11223344556677889
0.0.0.0/0	igw-12345678901234567
::/0	eigw-aabbccddee1122334

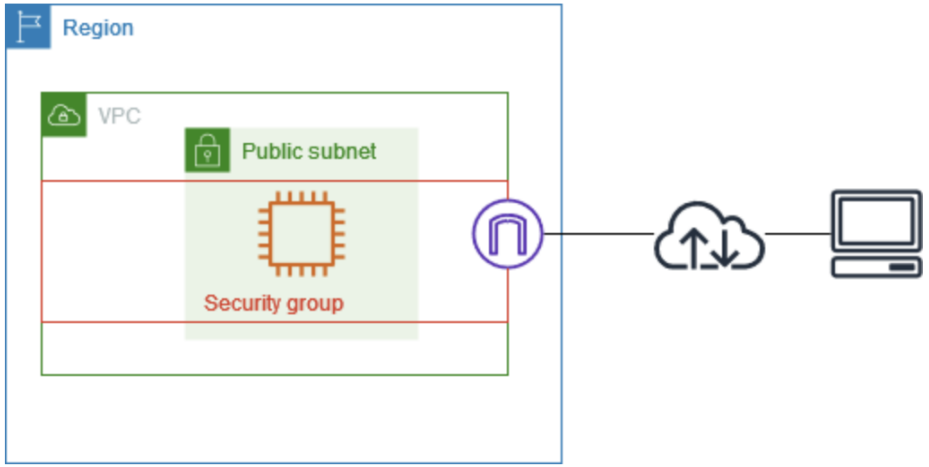
VPC Core Components

- Internet Gateways (IGWs): Allow communication between instances in VPC and the internet, enabling public subnets to send and receive traffic
- NAT Gateways: Allow instances in a private subnet to connect to services outside the VPC (like internet) without receiving inbound traffic.



VPC Security Layers

- Security Groups: Act as a virtual firewall to control inbound and outbound traffic at the associated resources like EC2.



Security Groups (SG)

Security groups are found on the **instance level**. They act as a **virtual firewall** that controls the traffic for one or more instances.

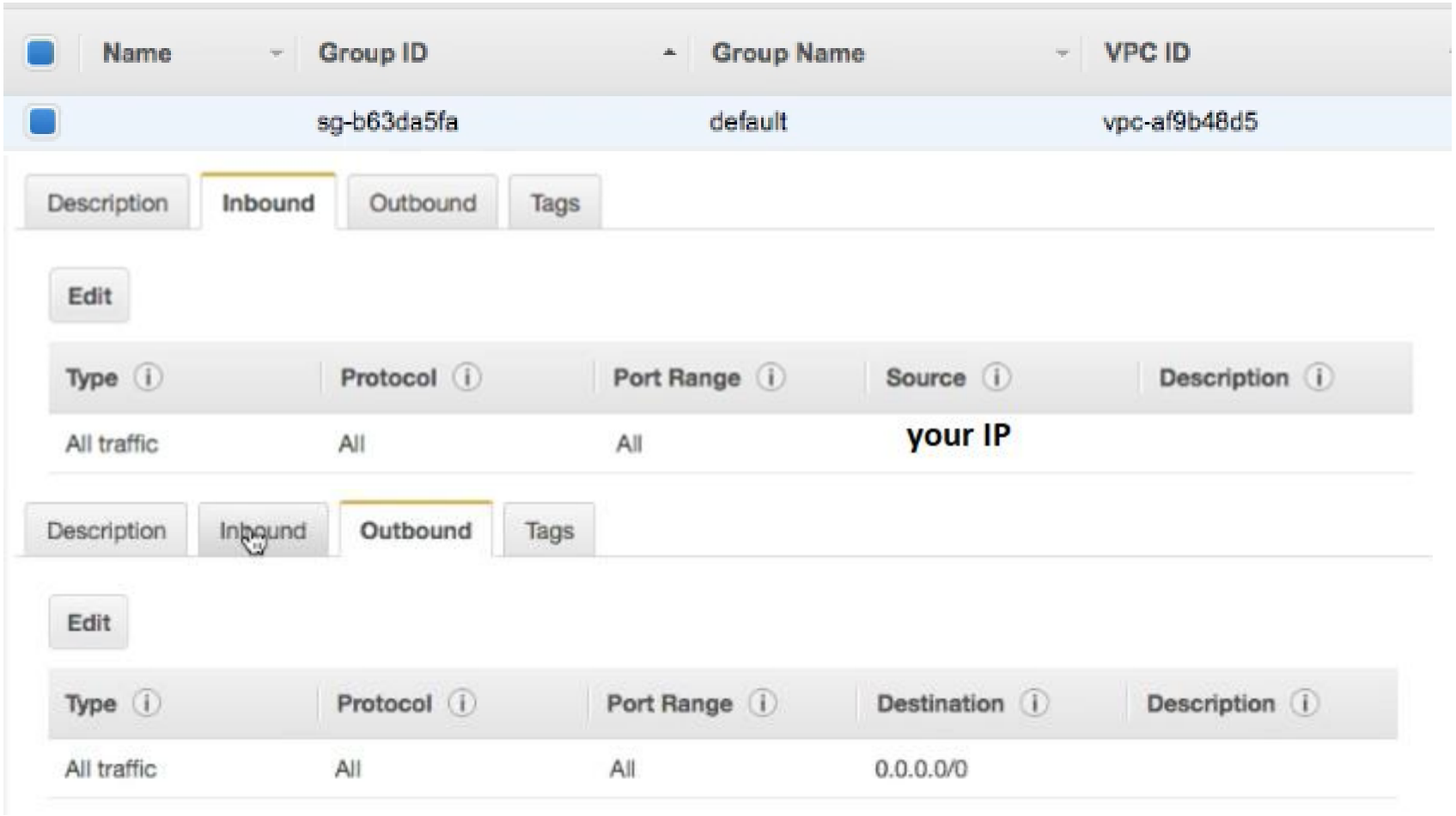
You add rules to security groups to **allow traffic** to (or from) its associated resource. By default, there is no inbound, which means everything is denied. You cannot write a deny rule like the NACL.

You can attach multiple SGs to an instance. All the rules from the security groups that are associated with the instance are evaluated.

By default, all outbound access is allowed. That is fine. We don’t touch the outbound rule of the SG.

Read more: Security Groups

Inbound/Outbound Rules



NACL

Inbound Outbound

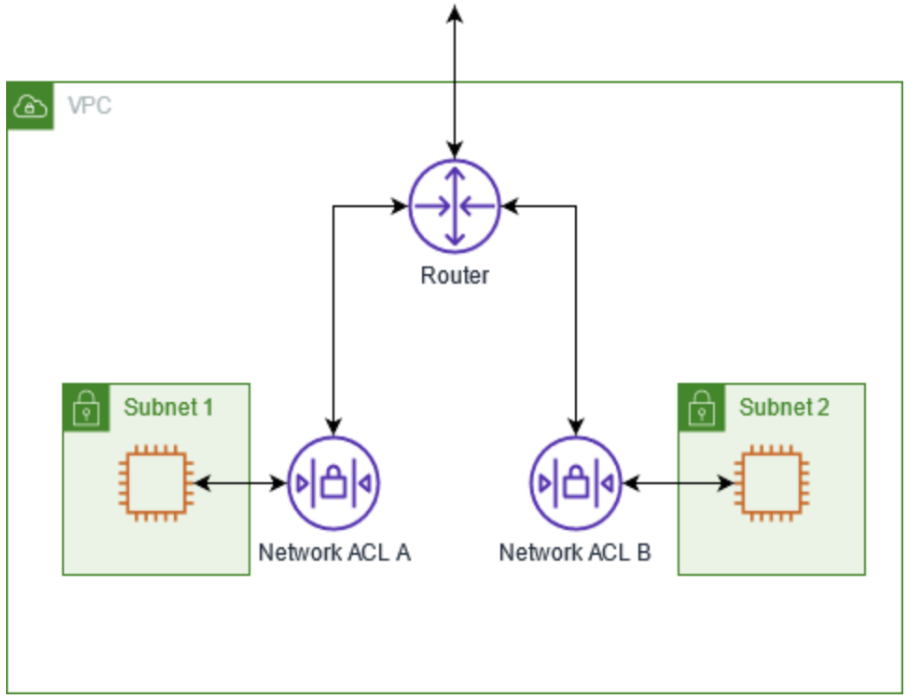
Security Group

EC2 Instance


VPC
NACL
Subnet
Security group I
EC2
Security group II
RDS

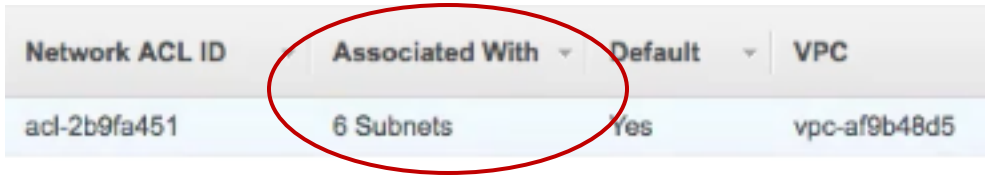
VPC Security Layers

- Network ACLs (NACL): Allows or denies specific inbound or outbound traffic at subnet level.



Network Access Control Lists - NACL

NACL acts as a firewall for controlling traffic in and out of one or more subnets. Your default VPC already has an NACL in place and is associated with all default subnets.



To access an EC2 instance from the internet, the request must pass the NACL security layer. If there is a rule that denies the request on the way, you cannot access the instance. Once it passed the NACL layer, then the request also has to pass Security Groups (SG) layer associated with the instance.

Read more about Network ACLs

NACL Rules

- The default NACL allows all traffic to the default subnets.
- Rules are evaluated from lowest to highest based on rule #. The first rule found that applies to the traffic type is immediately applied, regardless of any rules that come after it.
- The NACL allows or denies traffic from entering a subnet. Once inside the subnet, other AWS resources may have additional security layers (security groups).

Inbound

Rule #	Type	Protocol	Port Range	Source	Allow / Deny
100	ALL Traffic	ALL	ALL	0.0.0.0/0	ALLOW
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY

All traffic is allowed

Inbound

Rule #	Type	Protocol	Port Range	Source	Allow / Deny
90	SSH (22)	TCP (6)	22	0.0.0.0/0	DENY
100	SSH (22)	TCP (6)	22	0.0.0.0/0	ALLOW
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY

All traffic is denied

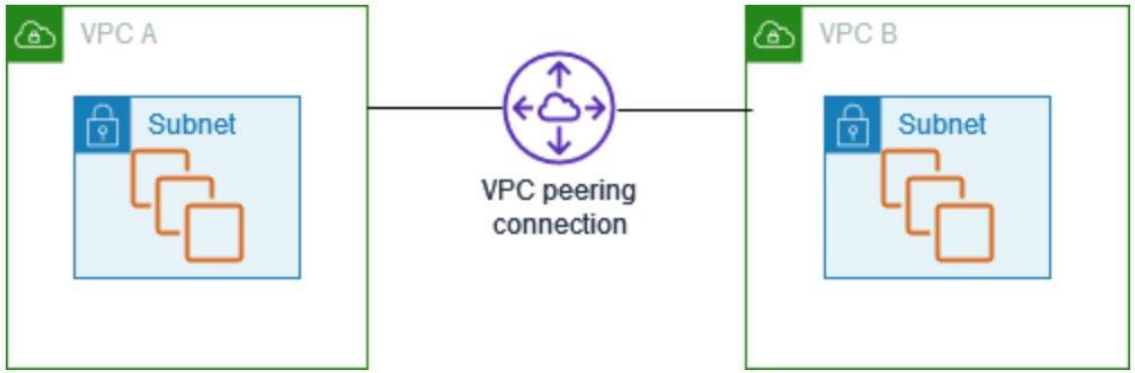
NACL	Security Group
Functions at **subnet** level.	Functions at **resource** level e.g., EC2 instances.
Stateless meaning any change applied to an incoming rule isn't automatically applied to an outgoing rule. Example: If a request comes through port 80, it should be explicitly indicated that its outgoing response would be the same port 80.	**Stateful**, meaning any changes which are applied to an incoming rule is automatically applied to a rule which is outgoing. Example: If the incoming port of a request is 80, the outgoing response of that request is also 80. The outbound is opened automatically.
NACL supports **allow** and **deny** rules.	SG supports **only allow rules**, and the default behavior is denial of all.
In case of NACL, the rules are applied in the order of their priority, wherein priority is indicated by the number the rule is assigned. This means every rule is evaluated based on the priority it has.	In the case of a security group, all the rules are applied to an instance. There is no priority.

VPC Peering and Transit Gateways

- VPC Peering is a network connection between two VPCs that enables to route traffic between them using private IP addresses.
- Transit Gateways: A service that simplifies the network architecture by allowing a single gateway to connect multiple VPCs, VPNs, and other networking services

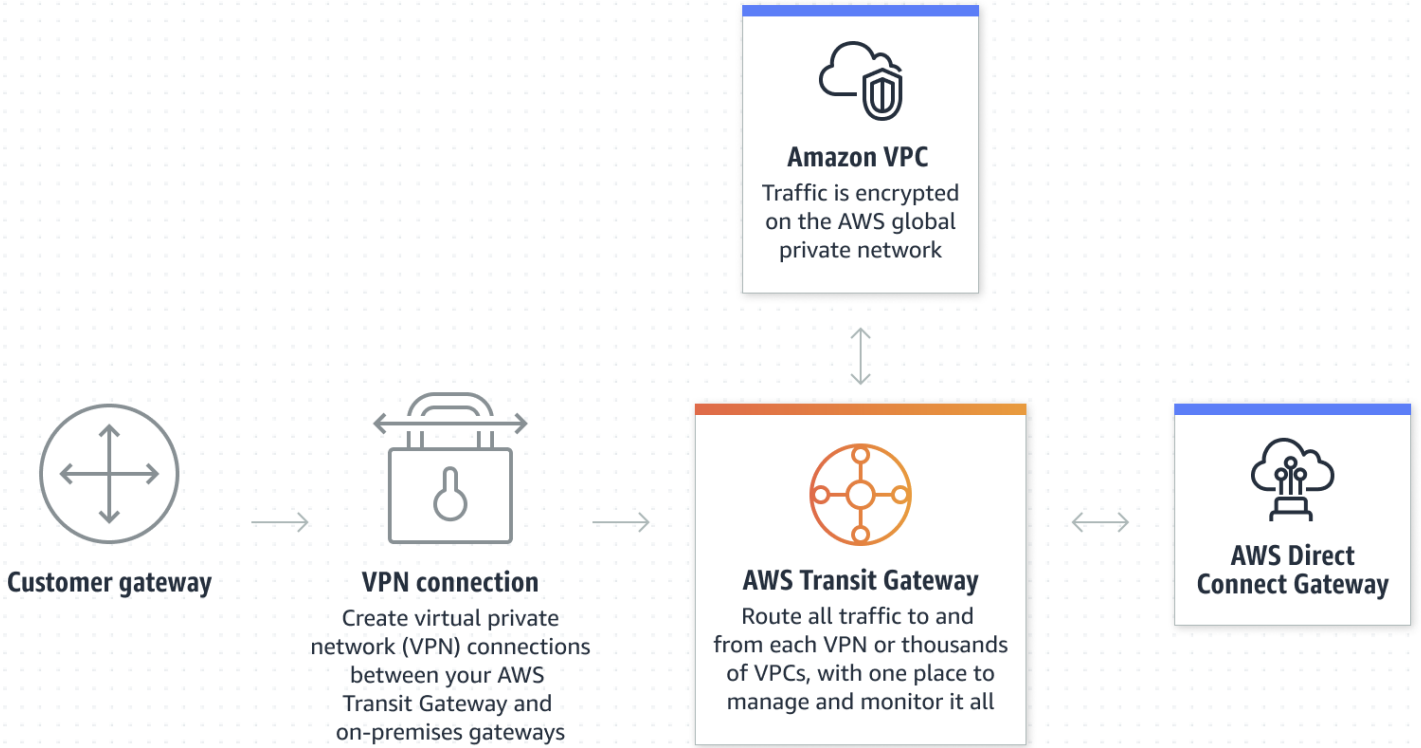
VPC Peering

- VPC Peering is a network connection between two VPCs that enables to route traffic between them using private IP addresses.



Transit Gateways

- A service that simplifies the network architecture by allowing a single gateway to connect multiple VPCs, VPNs, and other networking services

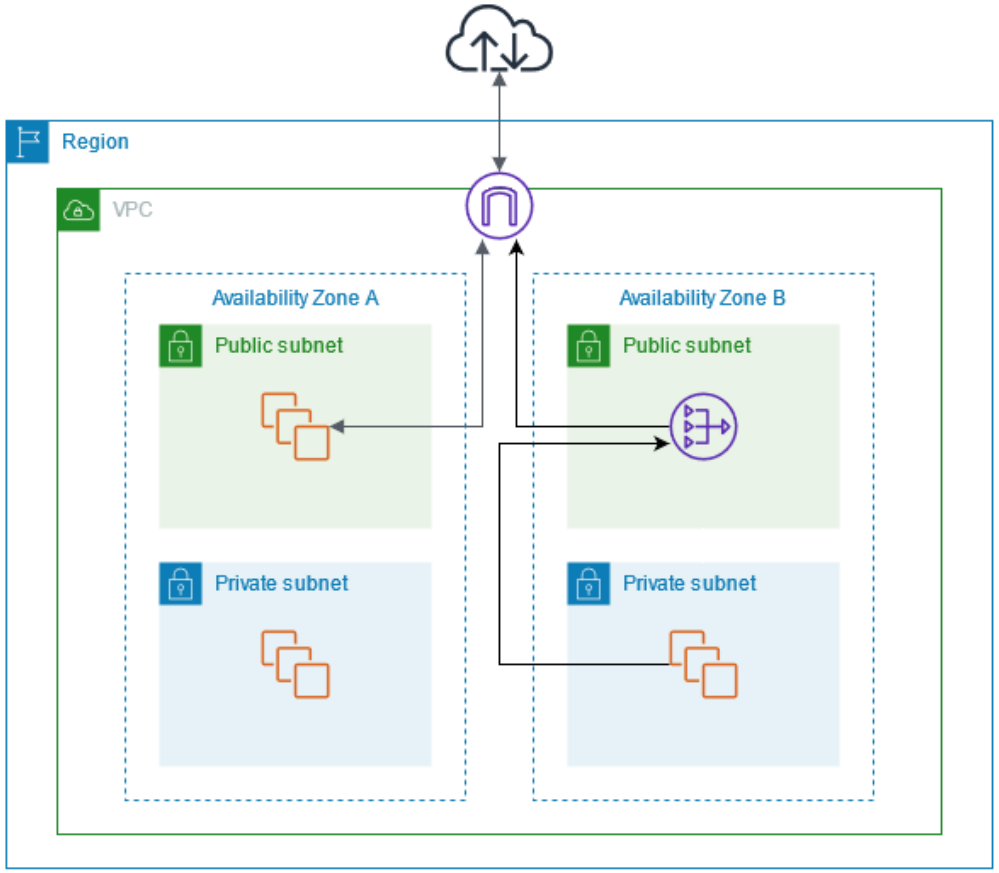


VPC Peering vs. Transit Gateways

- Choose the right architecture:
  - Network Complexity: Number of VPCs
  - Cross-region connectivity: Intra-region vs. Inter-region
  - Connection:
    - VPC Peering: No cost for connections
    - Transit Gateways: Fee per VPC attachment ($0.005)

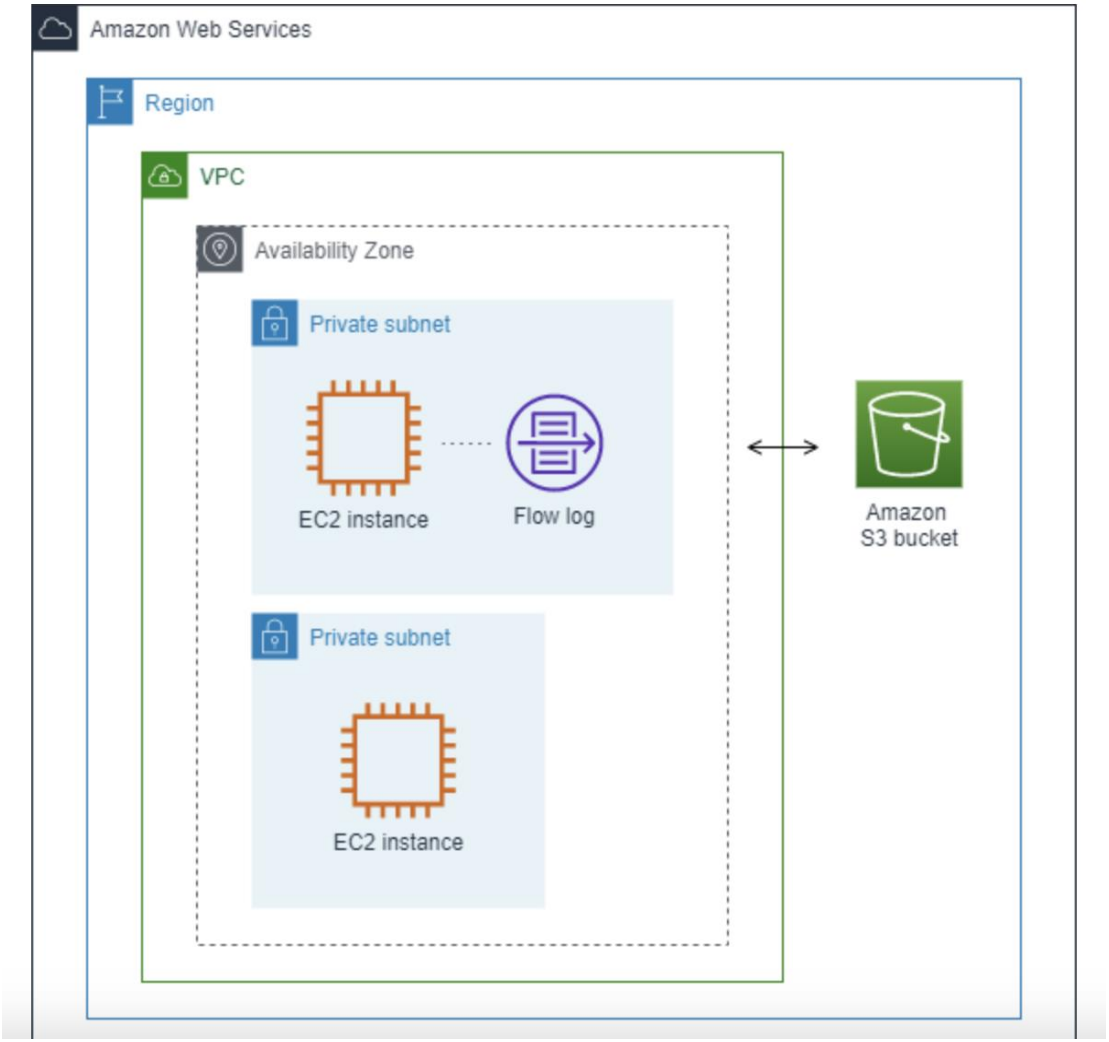
NAT Gateway

- You can use a public NAT gateway to enable instances in a private subnet to send outbound traffic to the internet, while preventing the internet from establishing connections to the instances.



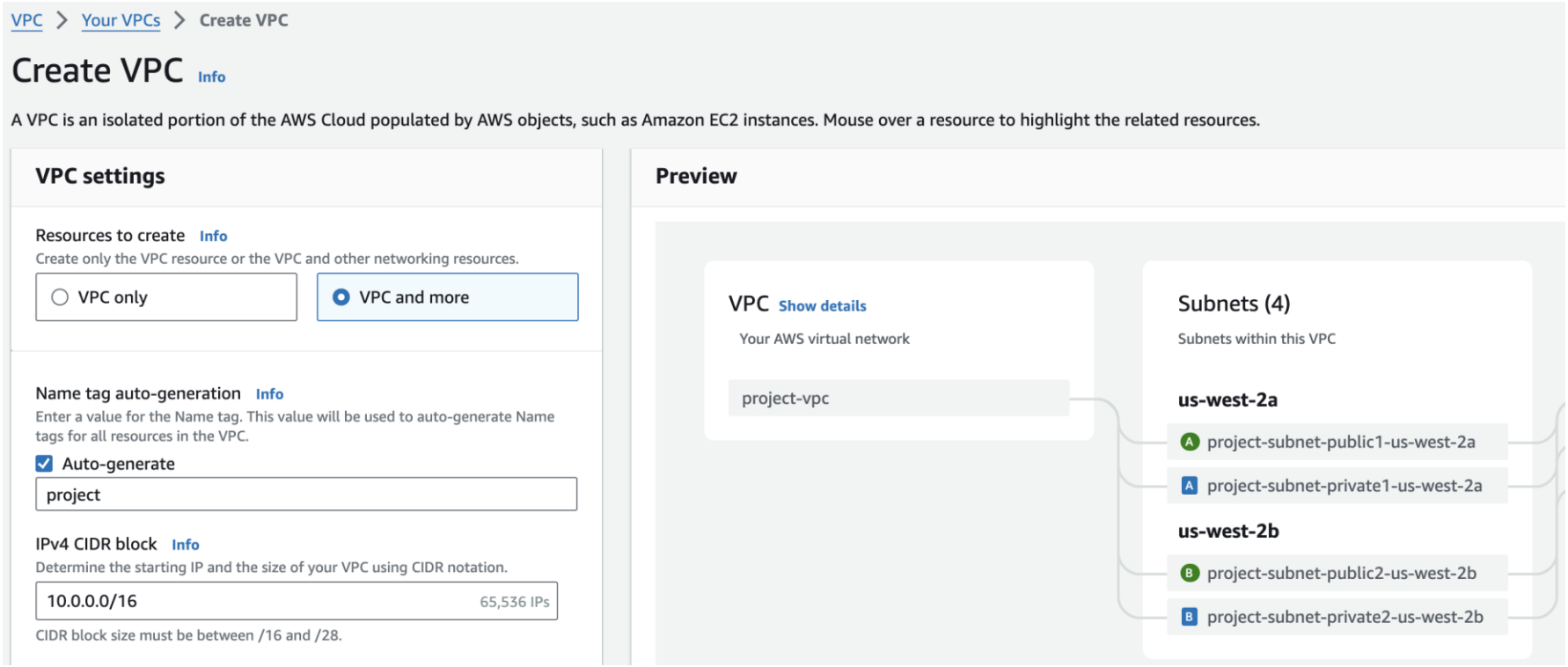
VPC Flow Logs

- Capture information about the IP traffic going to and from network interfaces in a VPC



VPC

- Demo


VPC > Your VPCs > Create VPC
Create VPC Info
A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.
VPC settings
Resources to create Info
Create only the VPC resource or the VPC and other networking resources.
VPC only
VPC and more
Name tag auto-generation Info
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.
Auto-generate
project
IPv4 CIDR block Info
Determine the starting IP and the size of your VPC using CIDR notation.
10.0.0.0/16
65,536 IPs
CIDR block size must be between /16 and /28.
Preview
VPC Show details
Your AWS virtual network
project-vpc
Subnets (4)
Subnets within this VPC
us-west-2a
project-subnet-public1-us-west-2a
project-subnet-private1-us-west-2a
us-west-2b
project-subnet-public2-us-west-2b
project-subnet-private2-us-west-2b

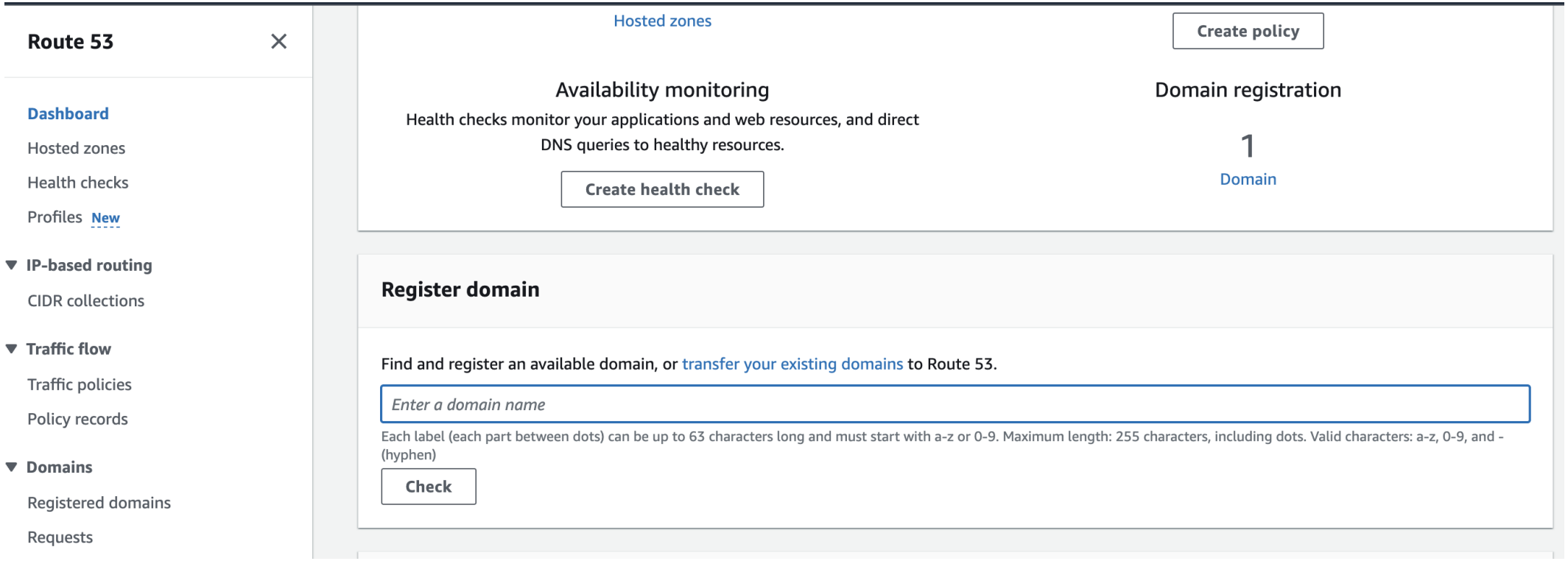
Amazon Route 53

- A highly available and scalable cloud Domain Name System (DNS) service designed to give developers and businesses an extremely reliable and cost-effective way to route end users to Internet applications.
- Connect user requests to infrastructure running in AWS, such as EC2, ELB, S3, etc.
- Can be used to route users to infrastructure outside of AWS.

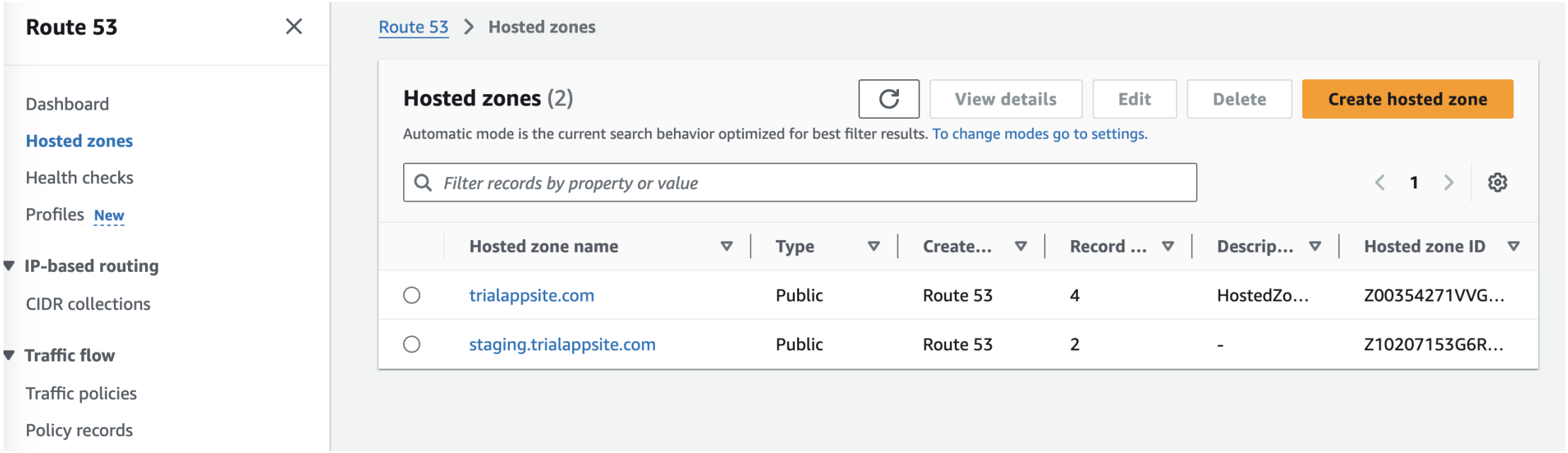
Amazon Route 53

- Domain Registration: Purchase and manage domains in Route 53
- DNS Service: Convert domain names into IP addresses
- Routing Policies: Determine how Route 53 responds to DNS queries.
- Health Check: Route 53 performs health checks on specified resources (e.g., web servers, email servers).
- DNS Failover: Increase the availability by redirecting traffic to healthy endpoints or to a static site if all endpoints fail.
- Hosted Zone: A container to manage the DNS records of a domain

Amazon Route 53



Route 53 - Hosted Zone



References

- https://docs.aws.amazon.com/
- ChatGPT: https://chatgpt.com/
- Google AI: https://gemini.google.com/app